

**p-value**

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- **$p$ -value:** The  $p$ -value is the observed “tail” probability of a statistics being at least as extreme as the particular observed value when  $H_0$  is true.
- If  $Y = u(X_1, \dots, X_n)$  is the statistics to be used in a test of  $H_0$  and if the critical region is of the form

$$u(x_1, \dots, x_n) \leq c,$$

an observed value  $u(x_1, \dots, x_n) = d$  would mean that the

$$p\text{-value} = \mathbb{P}(Y \leq d; H_0).$$

- If  $p$ -value is small, we need reject the null hypothesis.

- Let  $X_1, \dots, X_{25}$  be a random sample from  $N(\mu, 2^2)$ . To test

$$H_0 : \mu = 77 \quad \text{vs.} \quad H_1 : \mu < 77.$$

Say we observe the 25 values and determine that  $\bar{x} = 76.1$ . We know that  $Z = (\bar{X} - 77)/\sqrt{4/25}$  is  $N(0, 1)$  provided that  $\mu = 77$ . Since the observed statistics is  $z = (76.1 - 77)/0.4 = -2.25$ , the  $p$ -value of the test is  $\Phi(-2.25) = 1 - 0.998 = 0.012$ . Accordingly, if we use a significance level of  $\alpha = 0.05$ , we would reject  $H_0$  and accept  $\mu < 77$ .

- The p-value is the probability of observing data (or a test statistic value) as extreme as that observed or more so, if the null hypothesis is true.  
But that's not enough!
- Certain assumptions must be made to compute a p-value
  - An underlying statistical model (such as binomial, normal, linear, etc.)
  - Many other things related to that model (randomness or representativeness, no missing data, no response bias, and so on)
  - The null hypothesis for that model

- If [we assume]  $y_1, y_2, \dots, y_n$  are independent  $N(\mu, \sigma^2)$  and we perform a t test of  $H_0 : \mu = 0$ , a rejection could mean that
  - $\mu \neq 0$
  - or it could mean that the data are not independent
  - or it could mean that the data are not normal
  - or it could mean that the variances of the observations are not equal
- In other words, rejecting  $H_0$  suggests that something is wrong with the model. It does not specify what is wrong.

- If the p-value is “small enough,” the researcher concludes there is a “significant effect.”
- “Small enough” commonly means  $p < .05$ .
- If the p-value is small, this means that it is relatively unlikely that we would have seen the data we saw if all the assumptions were true.
- So, we either had bad luck (random error), or one or more of the assumptions may not be true.
- The null hypothesis assumption of no effect, is commonly THE assumption that is thought to be untrue.

- Wrong: The p-value is the probability that the null hypothesis is true.
  - Instead: The p-value indicates the degree to which the data are incompatible with what's expected if the null hypothesis and all the other assumptions used in the test are true.
- Wrong: The p-value is the probability that the observation (or data) occurred by chance.
  - Instead: The p-value is not the probability of chance given the data, it is the probability of the data given chance!

## More p-value misunderstandings

- Wrong: A significant test result ( $p < 0.05$ ) means that the null hypothesis is false and should be rejected.
  - Instead: A small P-value simply flags the data as being unusual if all the assumptions used to compute it (including the null hypothesis) were correct.
- Wrong: A non-significant test result ( $p > 0.05$ ) means that the null hypothesis is true or should be accepted.
  - Instead: A large P-value simply flags the data as not unusual if all the assumptions used to compute it (including the null hypothesis) were correct. Particularly tricky with small  $n$ , because only very extreme results would seem “unusual!” Perhaps none extreme enough!



## References

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- [1] *Introduction to Mathematical Statistics* by Hogg, McKean, Craig
- [2] *Probability and Statistics: The Science of Uncertainty* by Evans and Rosenthal
- [3] *A First Course in Probability* by Ross
- [4] *John E. Freund's Mathematical Statistics with Applications* by Irwin Miller Marylees Miller